ENTITY RELATIONSHIP DIAGRAM

Introduction

- Entity-Relationship diagram is used to visually represent data objects.
- The ER modeling technique is frequently used for the conceptual design of database applications and many database design tools employ its concepts.
- **Comparison with flowchart**
- Flow chart is a tool for designing a program; similarly ERD is a tool for designing databases.
- Also an ER diagram shows the kind and organization of the data that will be stored in the database in the same way a flowchart chose the way a program will run.

ER modelling objects

- In ER modeling, data is described as entities, relationships, and attributes

1. Entity:

- One of the basic components of ER model is entity.
- **An entity** is any distinguishable object about which information is stored. These objects can be person, place, thing, event or a concept.
- Entities contain descriptive information. Each entity is distinct.
- An entity may be **physical** or **abstract**. A person, a book, car, house, employee etc. are all **physical**
- entities whereas a company, job, or a university course, are **abstract entities**.
- It is normally represented by a rectangle shape.

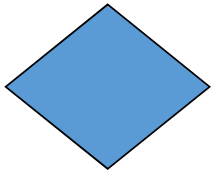


Entity Cont..d

- Entities are classified as **independent or dependent** (in some methodologies, the terms used are **strong and weak**, respectively).
- An **independent entity** is one, which does not rely on another entity for identification. A **dependent entity** is one that relies on another entity for identification.
- An **independent entity** exists on its own whereas **dependent entity** exists on the existence of some other entity.
- For example take an organization scenario. Here department is **independent entity**. Department manager is a dependent entity. It exists for existing depts. There won't be any department manager for which there is no dept.
- Some entity types may not have any key attributes of their own. These are called **weak entity types**.
- Entities belonging to a weak entity type are identified by being related to specific entities from another entity type in combination with some of their attribute values. For example, take the license entity. It can't exist unless it is related to a person entity.

2. *Entity Relationships*

- This is any association, linkage, or connection between the entities of interest to the business. Typically, a relationship is indicated by a verb connecting two or more entities.
- It is normally represented by a diamond shape.

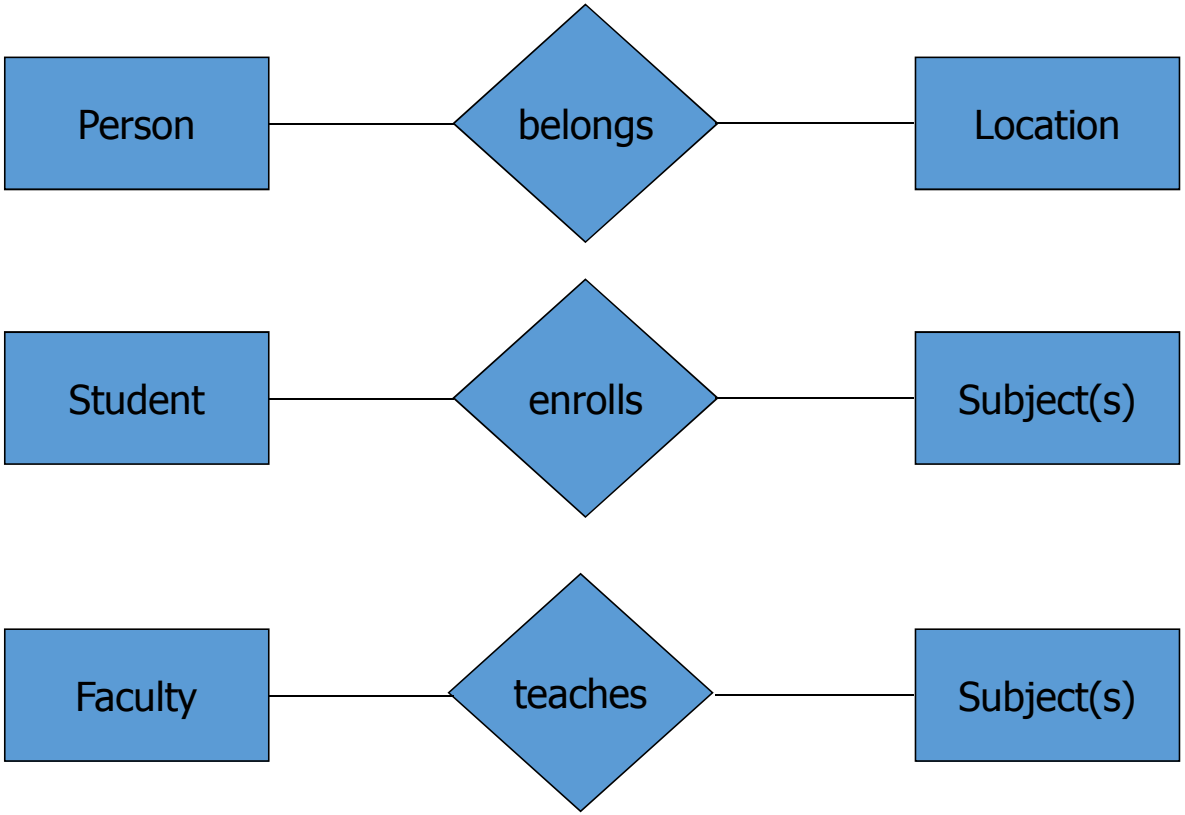


- Suppose there are two entities of our library system, member and book, then the relationship between them can be “borrows”.

i.e. Member borrows book

- Each relationship has a name, degree and cardinality.

Example of Entities with Relationship



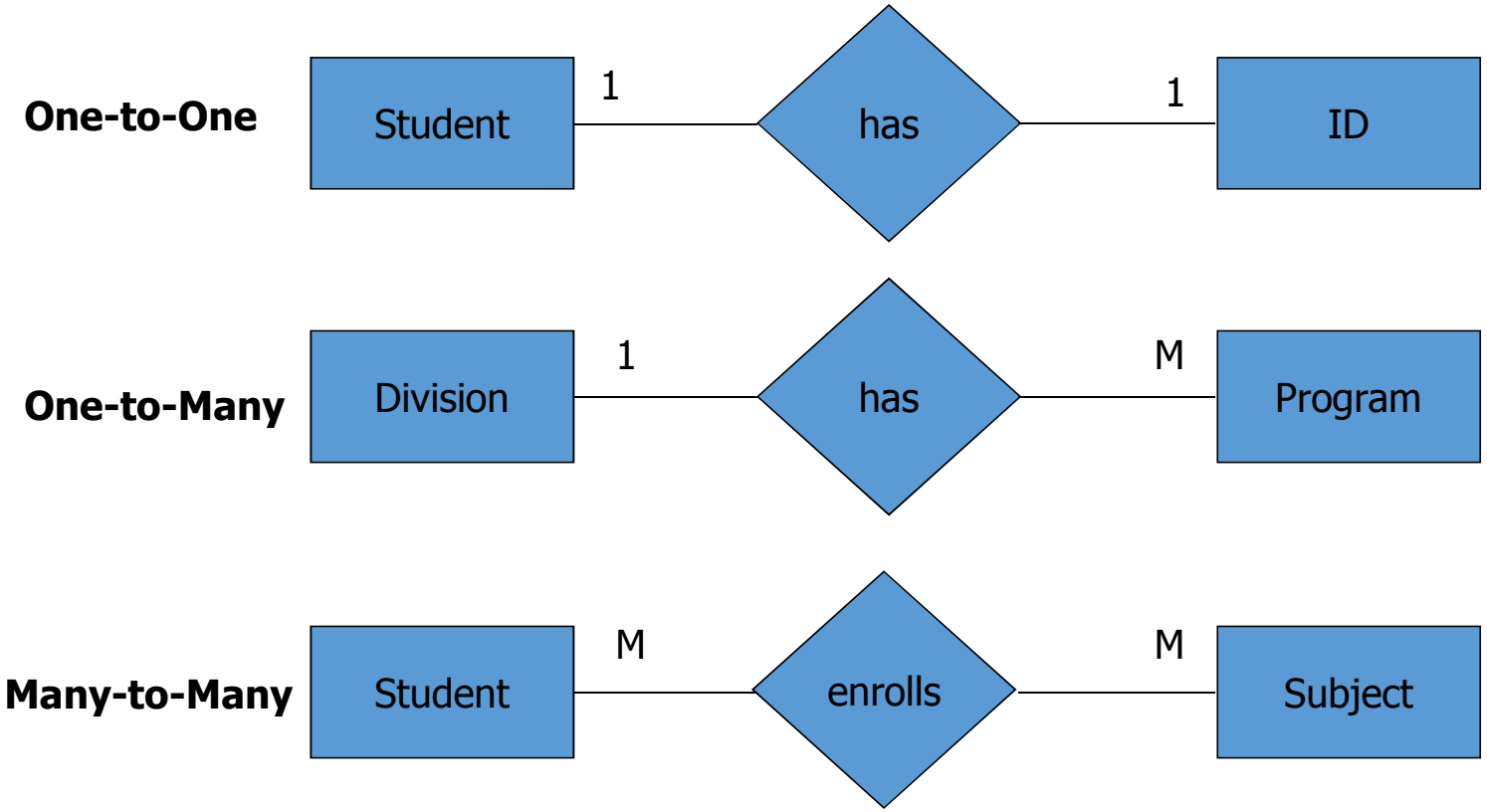
Connectivity and Cardinality

- **Connectivity** means how many instances of one entity are associated with how many instances of other entity in a relationship.
- **Cardinality** is used to specify such connectivity.
- The **connectivity of a relationship** describes the mapping of associated entity instances in the relationship.
- The **values of connectivity** are "one" or "many".
- The **cardinality of a relationship** is the actual number of related occurrences for each of the two entities.

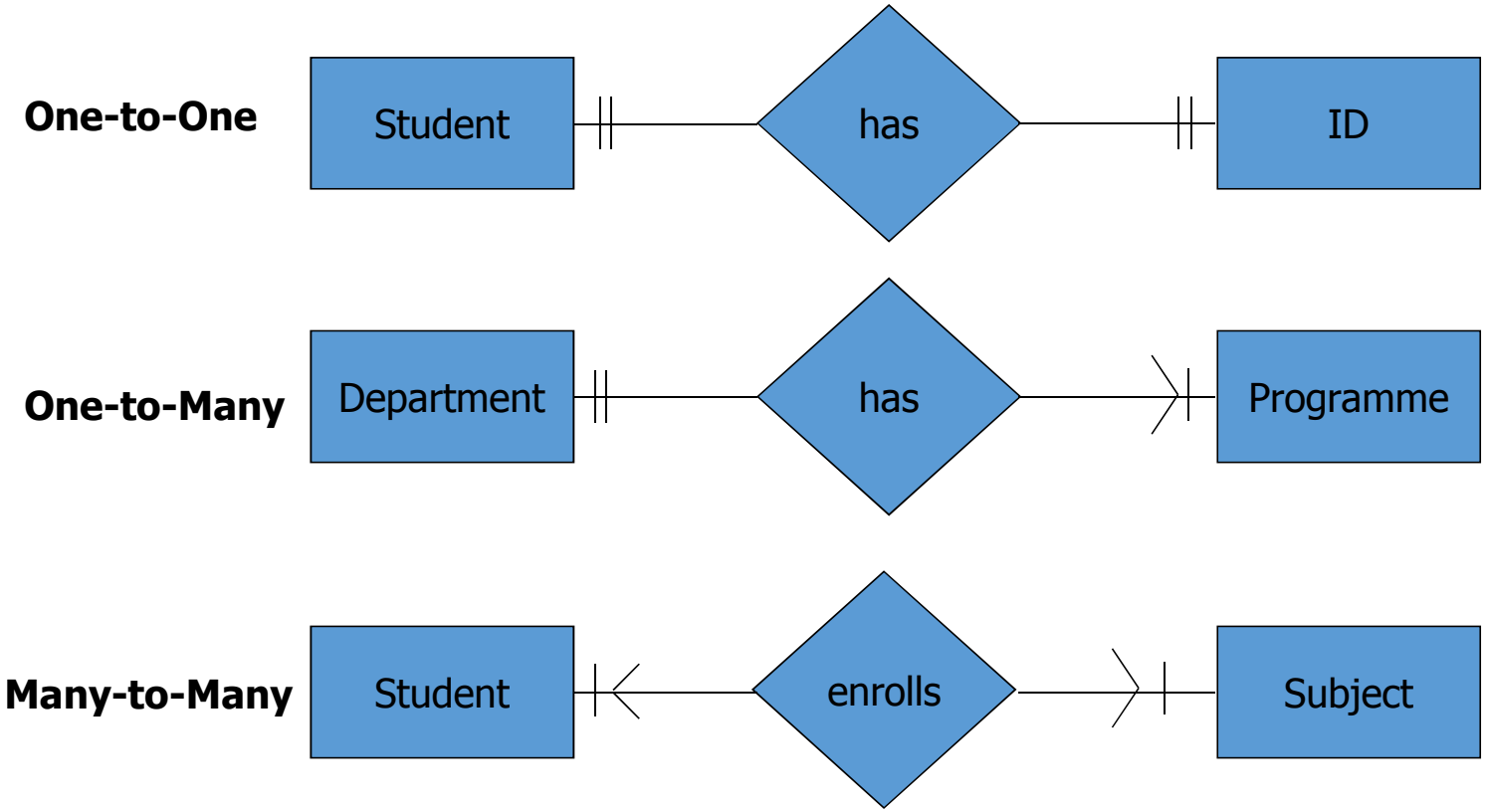
The basic types of connectivity for relations are:

i. one-to-one
ii. one-to-many
iii. many-to-many

Types of Relationship



Types of Relationship (Crow Notation)



Relationship types

- A **one-to-one (1:1)** relationship is when at most one instance of an entity A is associated with one instance of entity B.

  For example, each book in a library is issued to only one member at a particular time.

- A **one-to-many (1:N)** relationship is when for one instance of entity A, there are zero, one, or many instances of entity B but for one instance of entity B, there is only one instance of entity A.

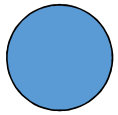
  An example of a 1:N relationships is a department has many employees; each employee is assigned to one department.

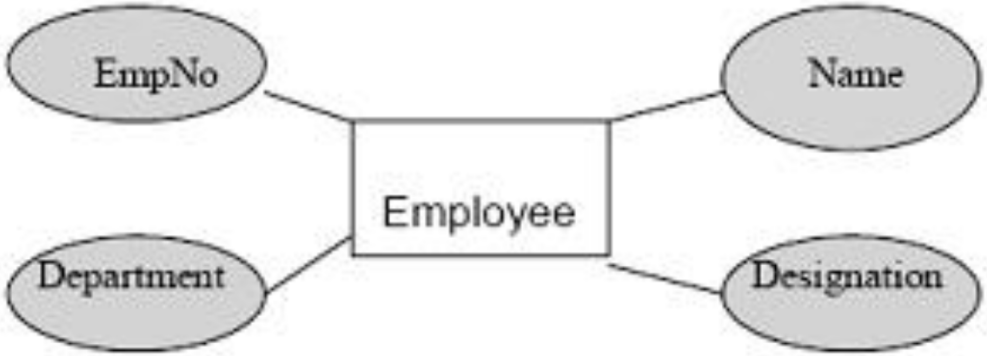
- A **many-to-many (M:N)** relationship, sometimes called non-specific, is when for one instance of entity A, there are zero, one, or many instances of entity B and for one instance of entity B there are zero, one, or many instances of entity A.

  An example is employees may be assigned to no more than three projects at a time; every project has at least two employees assigned to it.

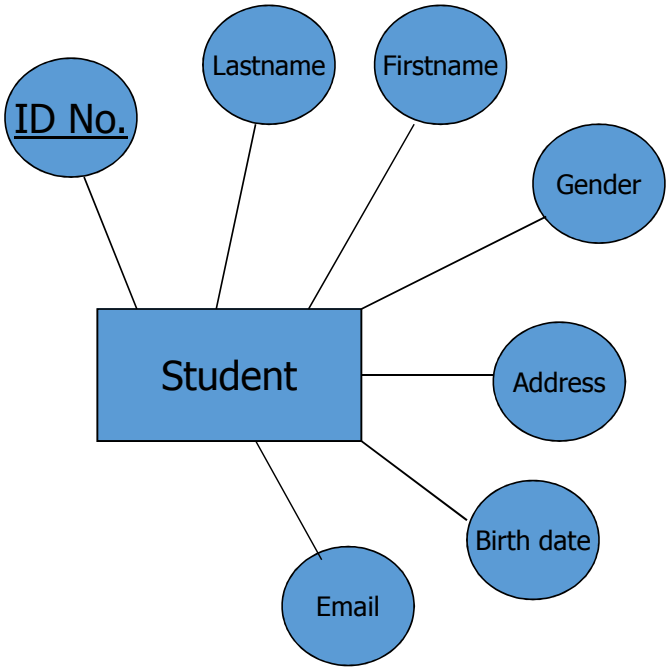
3. Attributes

- Attributes are basically properties of entity. We can use attributes for identifying and expressing entities.
- For example, Dept entity can have DeptName, DeptId, and DeptManager as its attributes.
- A Student entity could have the following **attributes** *ID Number*, *Last Name, First Name, Address, Birth Date* etc.
- A relationship could also have an attribute for example an Entity name **Student** enrolls (relationship) to a **Course/Program**. Now, when you enroll you enroll on a certain date so you will have an attribute of *Enrollment Date* under **Enroll** relationship.
- A car entity can have modelno, brandname, and color as its attributes.
- It is normally represented by a **circle**.



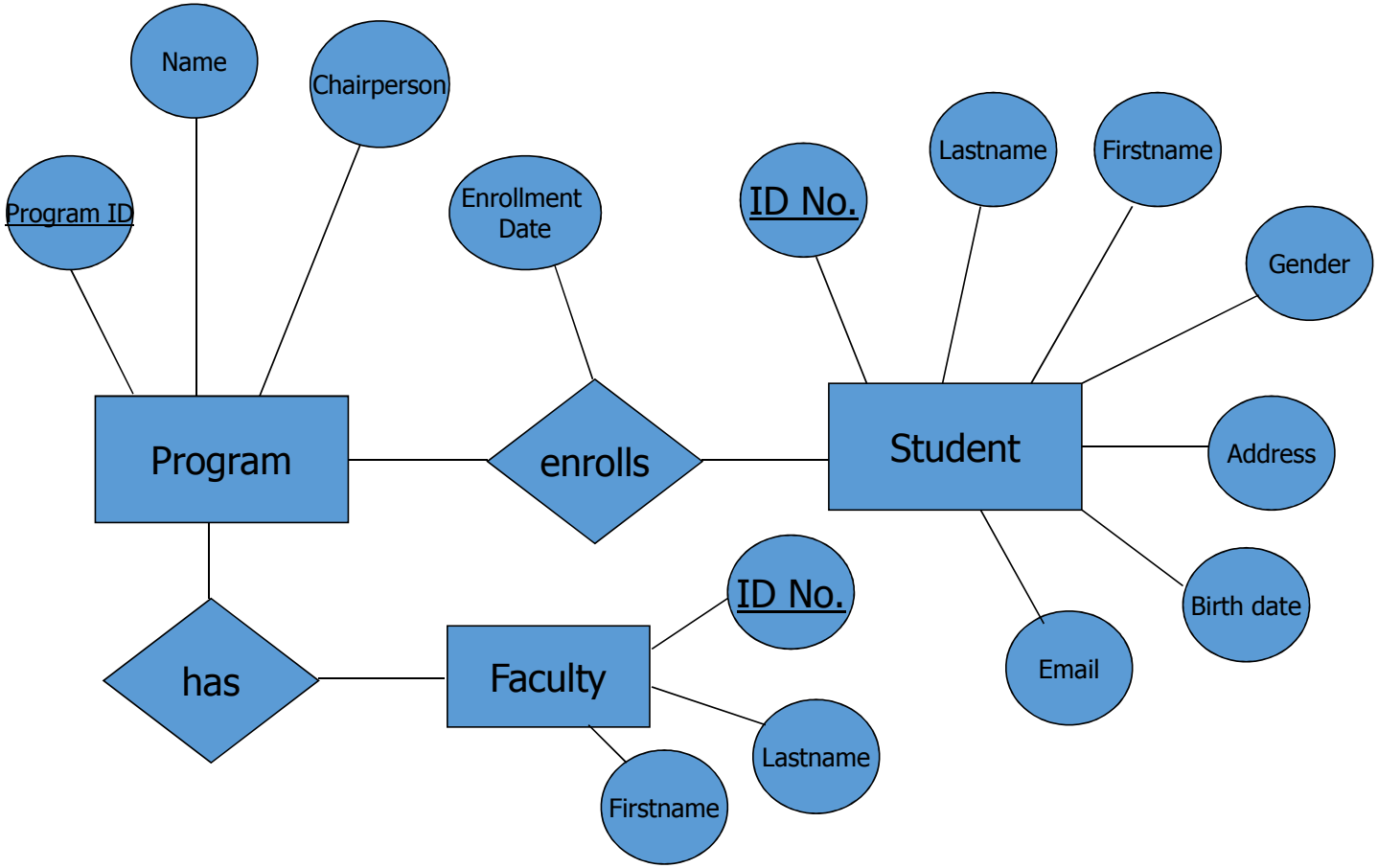


Example of Attributes

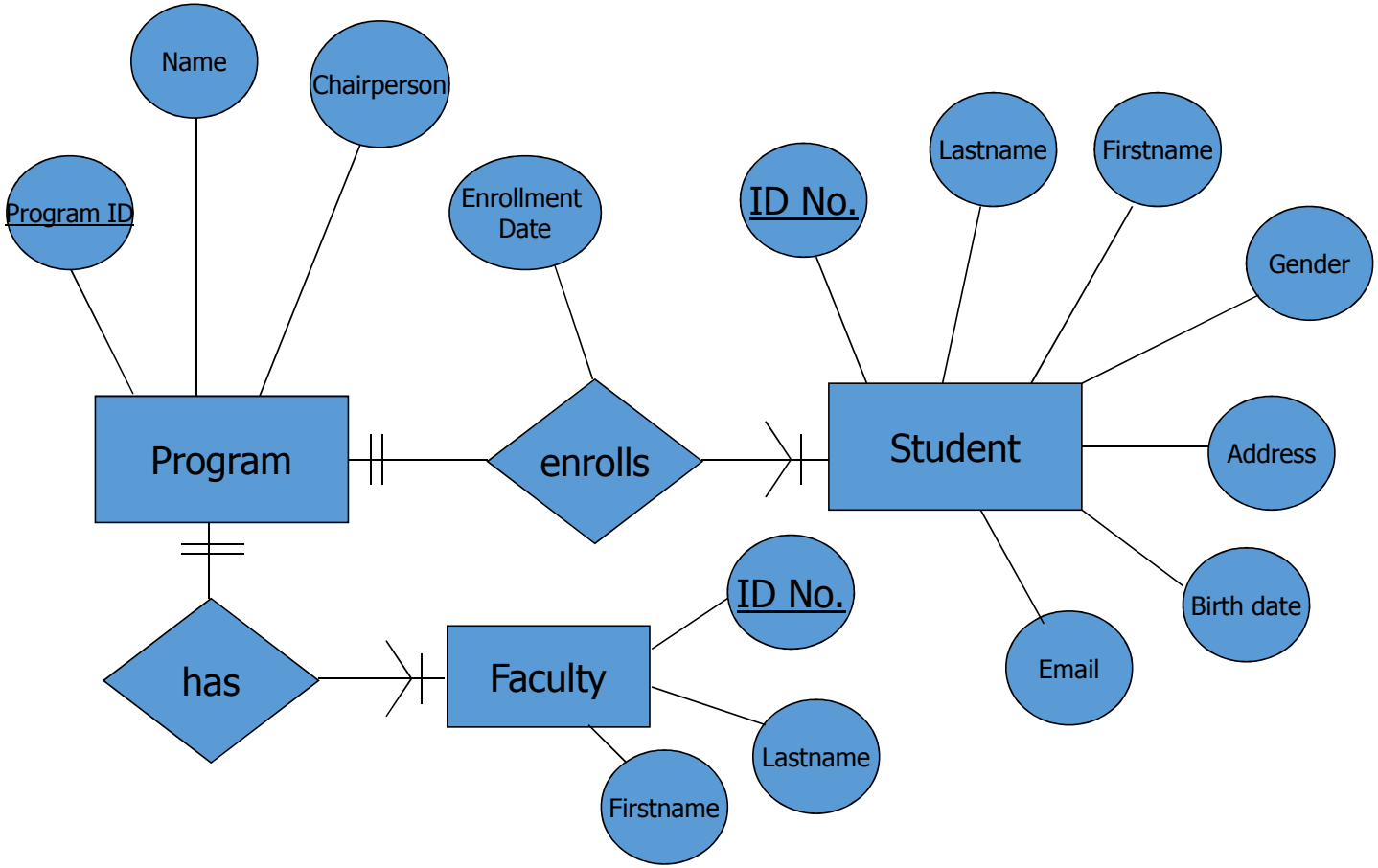


Note : A Primary Key is Underlined, in this case the ID No.

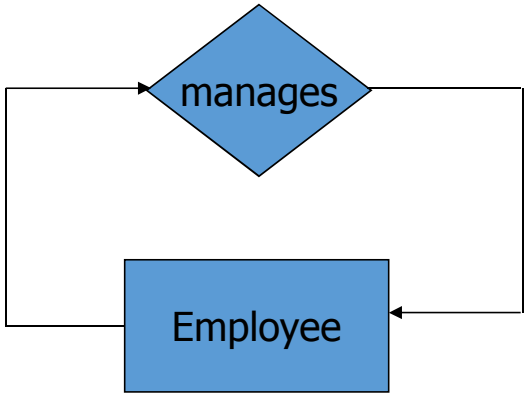
Example of Attributes



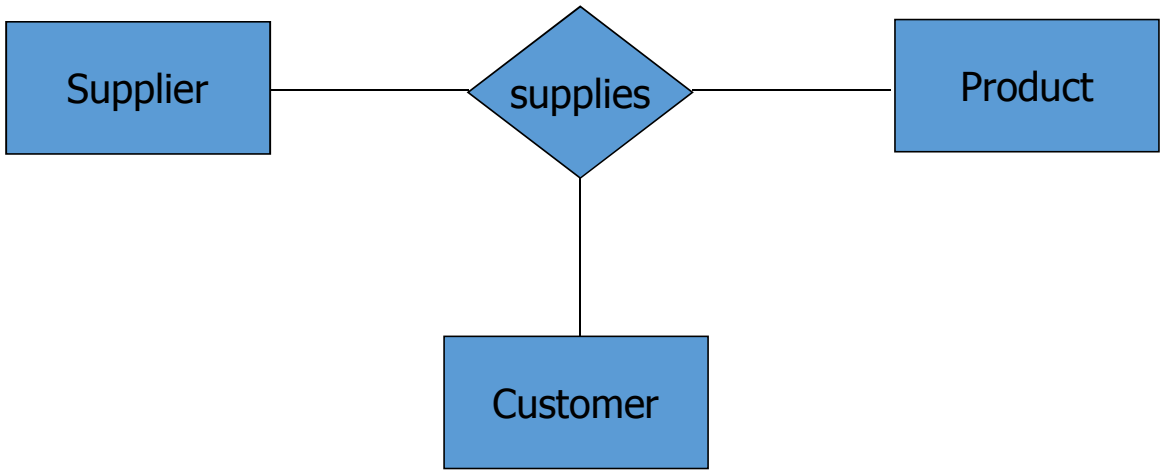
Example of Attributes



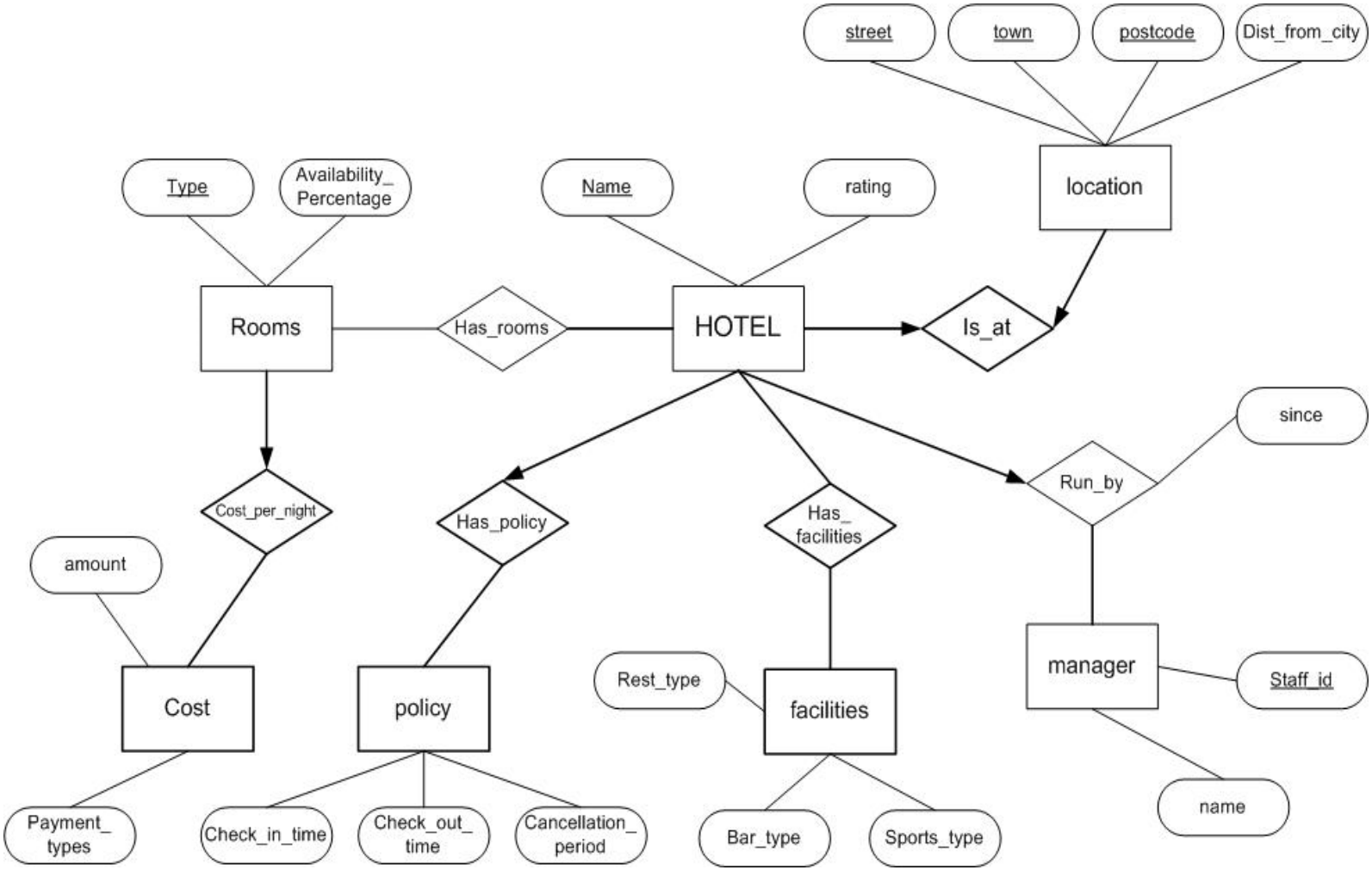
Recursive ERD



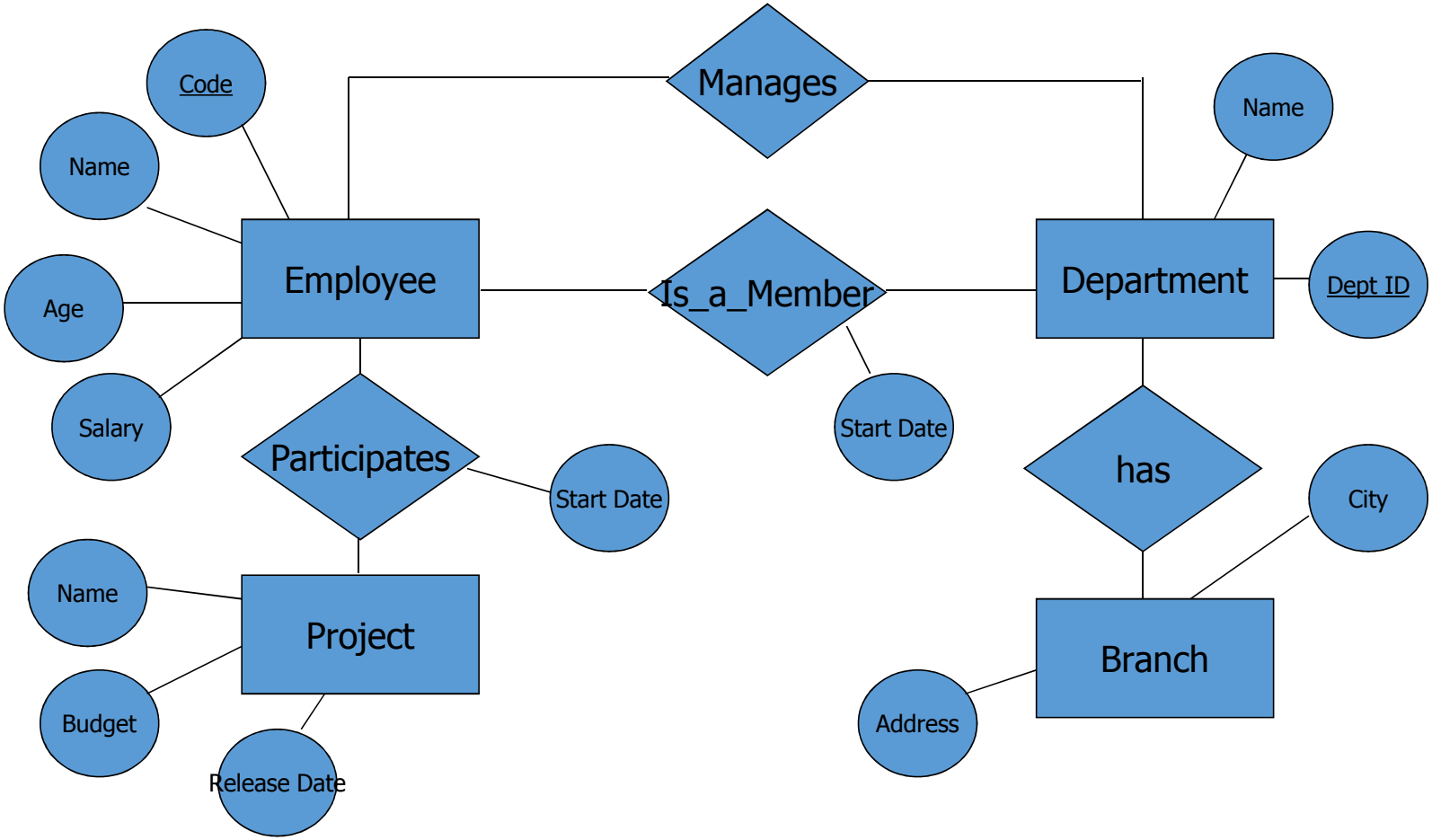
ERD with Ternary Relationship



ERD (More Examples)



ERD (More Examples)



Constructing ERD

- In order to begin constructing the basic model, the modeler must analyze the information gathered during the requirement analysis for the purpose of: and
- classifying data objects as either entities or attributes,
- identifying and defining relationships between entities,
- naming and defining identified entities, attributes, and relationships,
- documenting this information in the data document.
- Finally draw its ER diagram.

Example 1 of ERD

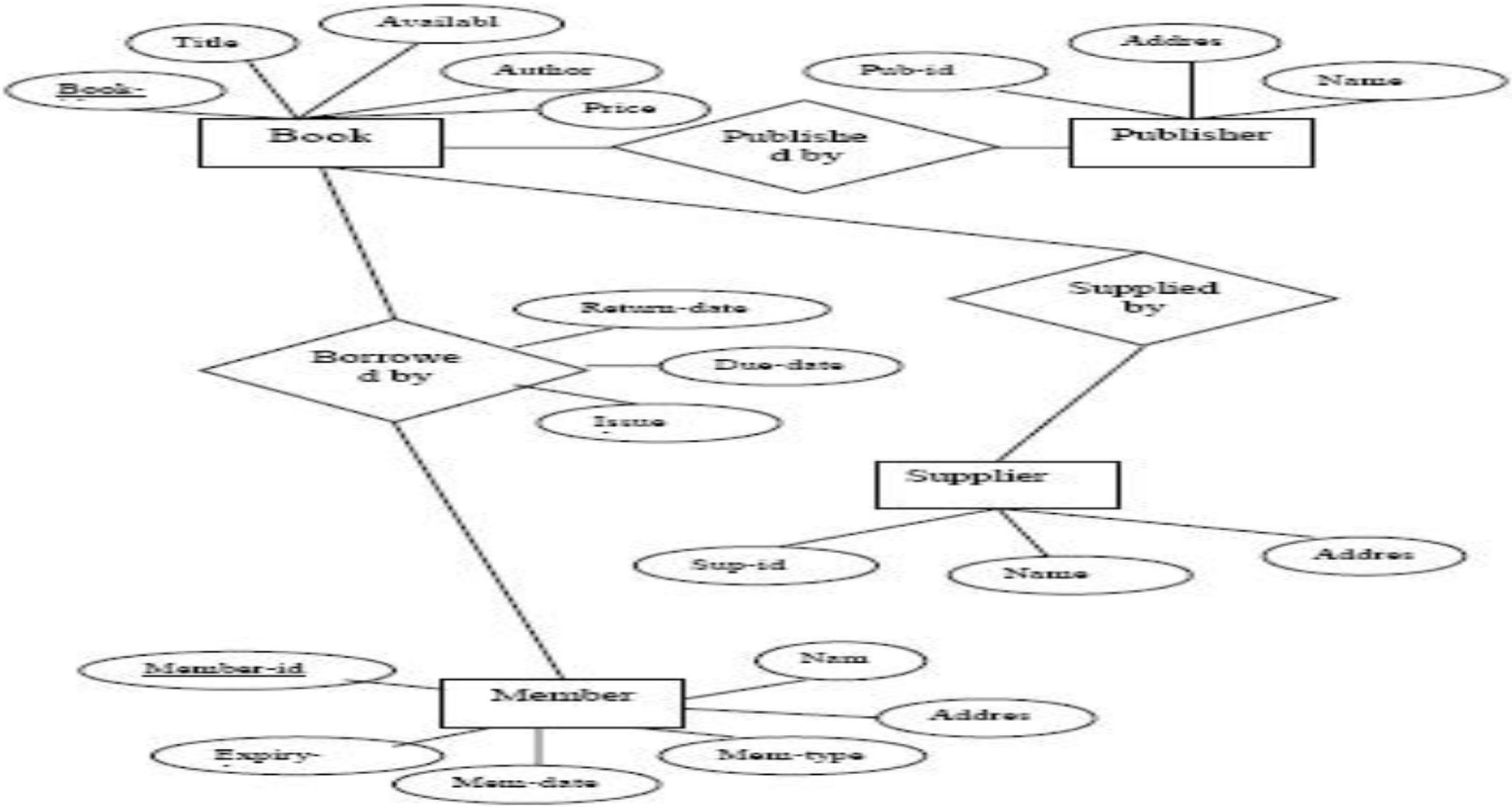
E-R Diagram for library management system

In the library Management system, the following entities and attributes can be identified.

- **Book** -the set all the books in the library. Each book has a Book-id, Title, Author, Price, and Available (y or n) as its attributes.
- **Member**-the set all the library members. The member is described by the attributes Member_id, Name, Street, City, Zip_code, Mem_type, Mem_date (date of membership), Expiry_date.
- **Publisher**-the set of all the publishers of the books. Attributes of this entity are Pub_id, Name, Street, City, and Zip_code.
- **Supplier**-the set of all the Suppliers of the books. Attributes of this entity are Sup_id, Name, Street, City, and Zip_code.

Assumptions: a publisher publishes a book. Supplier supplies book to library. Members borrow the book (only issue).

Example 1 of ERD



Example 2

- A country bus company owns a number of buses. Each bus is allocated to a particular route, although some routes may have several buses. Each route has a number of stages. Each bus goes through some of the towns. One or more drivers are allocated to each stage. Draw an entity model to show the corresponding entities and relationships

Example 2

